Cybersecurity - Undergraduate Level Assessment 25 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following is not an example of presentation layer issues?
   - (a) Poor handling of unexpected input can lead to the execution of arbitrary instructions
   - (b) Unintentional or ill-directed use of superficially supplied input
   - (c) Cryptographic flaws in the system may get exploited to evade privacy
   - (d) Weak or non-existent authentication mechanisms
2. ______________________ is the central node of 802.11 wireless operations.
   - (a) WPA
   - (b) Access Point
   - (c) WAP
   - (d) Access Port
3. Which of the following is not a block cipher operating mode?
   - (a) ECB
   - (b) CFB
   - (c) CBF
   - (d) CBC
4. When a wireless user authenticates to any AP, both of them go in the course of four-step authentication progression which is called ___________________
   - (a) AP-handshaking
   - (b) 4-way handshake
   - (c) 4-way connection
   - (d) wireless handshaking
5. A ________________ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
   - (a) front-door
   - (b) backdoor
   - (c) clickjacking
   - (d) key-logging

True / False (10 marks)

Answer True or False

6. True or False: A proxy firewall primarily operates at the physical layer, managing hardware connections and data transmission.
7. True or False: Public key encryption is faster than symmetric key cryptography, making it the preferred choice for all encryption needs.
8. True or False: A Base Signal Station is the term used for the device that connects mobile users to the network, similar to an Access Point.
9. True or False: A packet filter firewall operates at the network or transport layer of the OSI model to control data flow.
10. True or False: Merging tables can help prevent SQL injection by reducing the complexity of queries and limiting user input options.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find out the number of ways of painting the fence such that at most 2 adjacent posts have the same color for the given fence with n posts and k colors.
12. Write a function to remove the nested record from the given tuple.

End of Paper